AUTONOMOUS MAINTENANCE TRIAGE & VENDOR COORDINATION

More Doors, Same Team: Autonomous Maintenance Triage

A non-disruptive operational overlay that connects directly to your AppFolio inbox, clarifies tenant symptoms automatically, and prepares 1-tap vendor work orders before coordinators even start their day.

< 3 Min
Emergency Detection
Active water leaks, sparking breakers, and gas smells are trapped instantly and routed to on-call managers 24/7.

0 Min
Manual Data Entry
Unstructured tenant emails and portal texts are auto-formatted into structured AppFolio work orders with matched trade tags.

100%
Human Decision Control
Advisory-only architecture. No vendor dispatch or tenant commitment is ever finalized without 1-tap coordinator approval.

Operational Transformation: From Inbox Chaos to 1-Tap Dispatch

Assumed Baseline: 600-Door Scattered-Site Portfolio using AppFolio with 3 Property Coordinators

CAPACITY UNLOCKED: +40% DOORS / COORDINATOR

LEGACY FLOW
24–48 Hours Total

STEP 01
Tenant Submits
Vague email or portal note ("heater making noise, water dripping").
Unstructured

STEP 02
Inbox Sitting
Message waits unread overnight or during peak leasing hours.
8-12hr Delay

STEP 03
Email Ping-Pong
Coordinator chases tenant for missing photos, unit access, and pet notes.
3-4 Emails

STEP 04
Manual Re-Keying
Coordinator re-types details into AppFolio and searches vendor rates.
15 Min / Ticket

STEP 05
Vendor Chase
Phone calls or separate emails to schedule technician dispatch.
High Friction

AUTOMATED FLOW
Under 3 Minutes Total

01. INTAKE
Instant Webhook
Real-time ingestion from AppFolio API, email, or SMS listener.
< 2 Seconds

02. TRIAGE & AI
Safety & Trade Extraction
Deterministic rules check emergency hazards; AI structures trade scope.
Zero Delay

03. AUTO-CLARIFY
Missing Details Prompt
System auto-texts tenant for specific symptoms/photos if required.
Hands-Free

04. MANAGER DESK
1-Tap Mobile Decision
Coordinator reviews pre-matched vendor & scope on phone; taps Approve.
1 Tap to Dispatch

THREE UNCOMPROMISING PRODUCTION GUARDRAILS

Zero Silent Dispatch
The AI never sends emails to vendors or promises times to tenants independently. Every real action requires human 1-tap confirmation.

Zero Model Training
Tenant names, phone numbers, and lease notes are handled via strict enterprise APIs with zero data retention. Your operational data stays proprietary.

Hardcoded Safety Traps
Gas odors, sparking breakers, and active water leaks bypass language model interpretation entirely and trigger instant high-priority alerts.

END-TO-END EXECUTION PIPELINE

How It Works: From Tenant Message to Dispatched Vendor

A deterministic 5-stage pipeline running on your dedicated infrastructure—leaving your core AppFolio system of record 100% clean and perfectly audited.

STAGE 01 **Intake & Deduplication**	`AppFolio Webhook / Gmail IMAP` n8n Event Listener • 24/7 Polling	Captures incoming emails, SMS, and portal tickets. Checks **PostgreSQL database** for active threads within 24 hours to prevent duplicate work orders.	✓ Normalized Event Assigned ref: `MR-0184`
STAGE 02 **Classification & Safety Trap**	`gpt-5.6-luna (Structured JSON)` Hardcoded Safety Regex Engine	Extracts trade (Plumbing, HVAC, Electrical, Appliance), urgency tier (`EMERGENCY` vs `ROUTINE`), and distilled scope. Safety regex catches gas/water hazards immediately.	✓ Schema Validated Trade: `PLUMBING` | Sub: Leak
STAGE 03 **Autonomous Clarification**	`Automated SMS / Email Loop` Stateful Conversation Tracker	If critical details are missing (e.g. *"which bathroom?"*, *"is main shutoff closed?"*, *"pet on premises?"*), asks targeted follow-ups before routing to the manager.	✓ Ready For Review Context complete & verified
STAGE 04 **1-Tap Manager Decision**	`Telegram Bot / Slack / Mobile Push` Interactive Inline Action Cards	Manager receives a clean executive alert on their phone: tenant summary, matched preferred vendor with rate, and actionable buttons: **[Approve Dispatch]** or **[Hold for Review]**.	✓ Human Signed Off 1-Tap Approval Logged
STAGE 05 **Execution Lock & PMS Sync**	`PostgreSQL CTE Single-Send Lock` AppFolio REST API + Outbound SMTP	Database concurrency lock guarantees zero double-dispatch. Dispatches work order to preferred vendor with NTE budget, updates AppFolio record, and sends tenant confirmation.	✓ Closed & Synced Vendor emailed • AppFolio #4829

AUTOMATED HANDLING FOR CRITICAL EDGE CASES

EMERGENCY FAST-TRACK

Active Pipe Burst / Gas Odor

Immediately triggers emergency SMS + automated phone ring to on-call supervisor. Provides tenant with 1-tap utility shutoff guidance instantly.

NOISE REDUCTION

Rambling / Ambiguous Emails

Extracts the 2 core technical symptoms from 4 paragraphs of tenant prose. Prevents coordinator fatigue by surfacing clean technician summaries.

COMPLIANCE LOCK

Fair Housing / Legal Keywords

Rent withholding threats, mold disputes, or accommodation requests automatically pause triage and route to executive review with no automated drafting.

OPERATIONAL FIT & SYSTEMS ARCHITECTURE

Implementation Reality: What Changes vs. What Stays Untouched

Zero disruption to your daily operations. Your team keeps using AppFolio; the automation runs entirely as a background acceleration layer.

What Stays Untouched (Your Core)

ZERO DISRUPTION

- ✓ **AppFolio Property Manager:** Remains your primary system of record for leases, tenant records, work orders, and accounting.
- ✓ **Approved Vendor Relationships:** Your preferred plumbers, HVAC technicians, and electricians keep their existing rates and communication habits.
- ✓ **Manager Final Sign-Off:** Coordinators retain 100% control over whether a work order is dispatched or modified.

What We Add (The Middleware Layer)

AUTOMATED LAYER

- + **n8n Orchestration Core:** Runs 24/7 on an isolated, secure instance to process incoming webhooks and schedule tasks.
- + **PostgreSQL 16 Audit Database:** Stores idempotent message hashes, ticket states (`DRAFTED`, `ACKED`, `DISPATCHED`), and safety logs.
- + **Manager Mobile Dispatch Channel:** A secure Telegram or Slack bot delivering actionable approval cards straight to coordinators' phones.

DATA FLOW	SOURCE / CHANNEL	OPERATIONAL PURPOSE	PRIVACY & SECURITY
Tenant Intake Notes	`AppFolio / Email Webhook`	Symptom extraction, trade classification, and urgency scoring	✓ Zero Data Retention
Vendor Directory	`PMS Sync / Secure DB`	Matches closest trade specialist by property location & NTE budget	✓ Encrypted at Rest
Dispatch Authorization	`Manager Mobile Tap`	Single-send CTE lock triggers AppFolio sync and vendor notification	✓ Full Audit Log

Rapid 3-Week Staged Implementation Roadmap

LIVE WITHIN 21 DAYS

WEEK 01
Schema & Vendor Mapping
Connect AppFolio webhook/API, configure preferred vendor lists, trade categories, and NTE price thresholds.

WEEK 02
Shadow Mode Calibration
System processes live incoming messages in silent "shadow mode". Coordinators verify classification accuracy without dispatching.

WEEK 03
1-Tap Live Activation
Enable interactive mobile approvals for coordinators. Deliver full runbook, exception documentation, and team onboarding.

PRACTICAL NEXT STEP

15-Minute Technical Feasibility Alignment

If coordinator inbox fatigue or after-hours emergency response is a bottleneck in your operation, let's spend 15 minutes reviewing your current AppFolio setup and intake channels. We will confirm whether your volume justifies this automation—zero sales pitch, no commitment.

Book 15-Min Feasibility Call →

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